

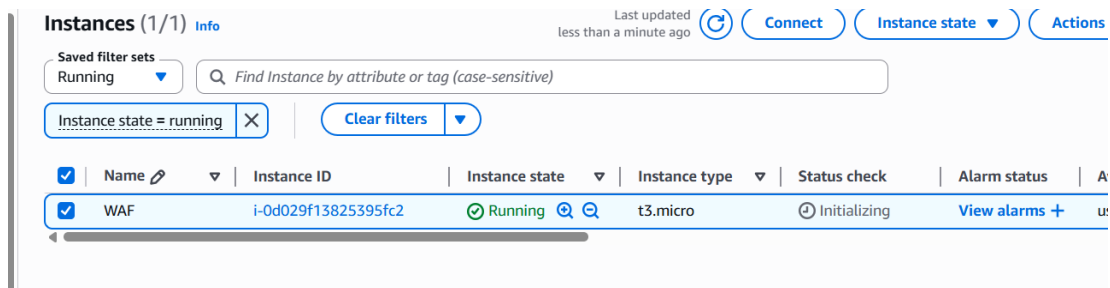
AWS WAF

AWS WAF (Web Application Firewall) lets you monitor and control HTTP/HTTPS requests forwarded to your web applications. It protects against common web exploits like SQL injection, cross-site scripting (XSS), bot traffic, and DDoS attacks at the application layer (Layer 7).

- Implemented AWS WAF integrated with an Application Load Balancer to protect an EC2-hosted Apache application against SQL Injection and malicious HTTP requests using AWS managed security rules and real-time traffic monitoring.

Implementations Steps:

1. Create a new EC2 Web Server and connect it



2. Install Apache Web Server

```
Last login: Fri May 8 07:12:32 2026 from 103.24.20.206
[ec2-user@ip-172-31-31-185 ~]$ sudo dnf install httpd -y
Amazon Linux 2023 Kernel Livepatch repository                259 kB/s | 34 kB    00:00
Dependencies resolved.

=====
Package                        Architecture Version                        Repository                Size
=====
Installing:
httpd                          x86_64      2.4.66-1.amzn2023.0.1         amazonlinux                47 k
Installing dependencies:
apr                            x86_64      1.7.5-1.amzn2023.0.4         amazonlinux                129 k
apr-util                       x86_64      1.6.3-1.amzn2023.0.2         amazonlinux                97 k
apr-util-ldap                  x86_64      1.6.3-1.amzn2023.0.2         amazonlinux                13 k
generic-logos-httpd            noarch      18.0.0-12.amzn2023.0.3       amazonlinux                19 k
httpd-core                     x86_64      2.4.66-1.amzn2023.0.1         amazonlinux               1.4 M
httpd-filesystem               noarch      2.4.66-1.amzn2023.0.1         amazonlinux                13 k
httpd-tools                    x86_64      2.4.66-1.amzn2023.0.1         amazonlinux                81 k
libbrotli                      x86_64      1.0.9-4.amzn2023.0.2         amazonlinux               315 k
mailcap                         noarch      2.1.49-3.amzn2023.0.3       amazonlinux                33 k
Installing weak dependencies:
apr-util-openssl               x86_64      1.6.3-1.amzn2023.0.2         amazonlinux                15 k
mod_http2                      x86_64      2.0.27-1.amzn2023.0.3       amazonlinux               166 k
mod_lua                        x86_64      2.4.66-1.amzn2023.0.1         amazonlinux                60 k
```

- Installed the Apache web server and run it

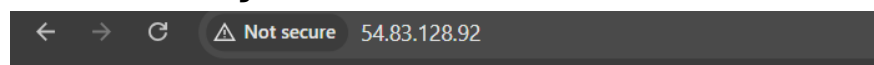
```
mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-31-185 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-31-185 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-31-185 ~]$ sudo systemctl enable httpd
[ec2-user@ip-172-31-31-185 ~]$
```

3. Created a new sample web page and verify it

```
[ec2-user@ip-172-31-31-185 ~]$ sudo systemctl enable httpd
[ec2-user@ip-172-31-31-185 ~]$ echo "WAF Demo Application Running" | sudo tee /var/www/html/index.html
WAF Demo Application Running
[ec2-user@ip-172-31-31-185 ~]$
```

Successfully Done



WAF Demo Application Running

4. Now Create Application Load Balancer

[EC2](#) > [Load balancers](#) > Create Application Load Balancer

Create Application Load Balancer [Info](#)

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers. When a connection request arrives, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group.

► How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

WAF-ALB

• Listeners

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80 Remove

Protocol: HTTP Port: 80
1-65535

Default action [Info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

• Security and VPC default

Security groups

Info

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected outbound rule to allow traffic to your load balancer.

Security groups

Select up to 5 security groups

default (sg-04eeb95c40a6368d8)

X

Inbound rule (12)

Outbound rule (2)

5. Created Target Group

EC2 > Target groups > WAF-TG

board

Global View

is

ances

nces

nce Types

ch Templates

Requests

gs Plans

ved Instances

WAF-TG

Successfully created the target group: WAF-TG. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the Targets tab.

Give feedback

Details

arn:aws:elasticloadbalancing:us-east-1:928430096503:targetgroup/WAF-TG/6d7777c33a7a7073

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

vpc-096ec8a65d98c6729

IP address type

IPv4

Load balancer

None associated

6. Successfully created Application load balancer

EC2 > Load balancers > WAF-ALB

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

WAF-ALB

Successfully created load balancer: WAF-ALB

It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

Actions

Details

Load balancer type

Application

Status

Provisioning

VPC

vpc-096ec8a65d98c6729

Load balancer IP address type

IPv4

Scheme

Internet-facing

Hosted zone

Z35SXDOTRQ7X7K

Availability Zones

subnet-05892b6b6d9730f66 us-east-1a (use 1-az1)

subnet-0a25ef5df35935262 us-east-1c (use 1-az4)

Date created

May 8, 2026, 13:20 (UTC+05:30)

7. Verified ALB Working

← → ↺

Not secure waf-alb-648571988.us-east-1.elb.amazonaws.com

WAF Demo Application Running

8. Create Protection for Web ACL

Create protection pack (web ACL)

Tell us about your app [Info](#)

Sharing your app category and app focus detail helps AWS recommend the best security protection.

App category

Select one or more app categories

Enterprise & business applications X

Select at least one app category.

App focus

☒ **Both API and web**
 App engages with users through both a website or mobile application along with programmatic means.

☐ **API**
 App engages with users primarily through programmatic means.

☐ **Web**
 App engages with users primarily through a website or mobile application interface.

Select resources to protect (1/1) [Info](#)

Remove Add resources

Search for a resource by its name or type

☒	Name	Type
☒	WAF-ALB	Application Load Balancer

Initial Protections chosen Essential Rules

Choose initial protections [Info](#)

AWS WAF offers various protection packages of WAF rules. AWS WAF rule configurations are based on security best practices. You should validate the rule configuration. You can also choose individual rules instead of packages.

Recommended

☐ **Recommended rules for you**
Enables recommended rules for the selected application categories and app focus.

Everything in Essentials and the following

- ☒ Rate limit for GET requests Count
- ☒ Rate limit for POST, PUT, DELETE requests Count
- ☒ Amazon IP reputation list Default
- ☒ Anonymous IP list protection Default
- ☒ Managed IP DDoS protection Default
- ☒ Bot Control protection Count

Essentials

☒ **Essential rules**
Enables essential rules for the selected application categories and app focus.

- ☒ Layer 7 anti-DDoS Count
- ☒ IP allowlist Allow
- ☒ IP blocklist Block
- ☒ Geographic restriction Block
- ☒ Global rate limit Count
- ☒ Body size restriction rule Count
- ☒ AWS core rule set Default
- ☒ Known bad inputs protection Default
- ☒ SQL injection protection Default

You build it

☐ **Build your own pack from protections AWS WAF off**
Select and customize rules to

☒ You choose which rules to apply

Given Name

Name and describe

Name

Used to identify this app protection and its CloudWatch metrics

WAF-ACL

Name **cannot** be changed later. Valid characters are letters, numbers, and hyphens (-).

Description - optional

Enter a description to help identify this protection pack (web ACL) later

The description can have up to 256 characters.

9. Created Web ACL

Protection packs (web ACLs) (1) [Info](#) [Actions](#) [Create protection pack \(web ACL\)](#)

Use protection packs (web ACLs) to combine rules and rule sets that protect your infrastructure from threats.

Search protection packs (web ACLs) by name, i Region scope: CloudFront (Global) and Regional < 1 > [List](#) [Grid](#)

Name	Rules	Resources	Dashboard	AI traffic analysis new	Sampled requ...	Logging & metrics	Scope
WAF-ACL	12 rules	Manage	View	View	View	Not enabled	Regional

WAF & Shield > Protection packs (web ACLs) > WAF-ACL



Smart recommendations

Based on traffic potential vulnerability, and aggregate data, we will give you tailor made recommendations to stop malicious actors before they attack.

WAF-ACL [JSON](#)

ARN: [arn:aws:wafv2:us-east-1...](#) Id: [eb74f162-a12c-48ee...](#)

- Manage details
- Manage rules
- Manage custom response bodies
- Manage resources
- View dashboard, logs and sampled requests
- View AI traffic analysis [new](#)
- Configure logging and sampled requests

Protection packs (web ACLs) (1) [Info](#) [Actions](#) [Create protection pack \(web ACL\)](#)

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Name	Rules	Resources	Dashboard	AI traffic analysis new	Sampled requ...	Logging & metrics	Scope
WAF-ACL	12 rules	Manage	View	View	View	Not enabled	Regional

- Open Managed Resources
- Check ALB attached or not

Manage resources [X](#)

[WAF-ACL](#)

WAF-ALB Application Load Balancer

Resource level DDoS protection

Protect critical resources from DDoS attacks using AWS managed threat list. This configuration currently only applies to Application Load Balancers.

Protection against low reputation sources Active under DDoS [Pencil icon](#)

[Cancel](#) [Disassociate](#) [Add regional resources](#)

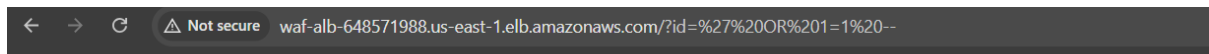
web ACL)

10. Check DNS of load balancer still active or not

[←](#) [→](#) [↻](#) [⚠ Not secure](#) waf-alb-648571988.us-east-1.elb.amazonaws.com

WAF Demo Application Running

11. Now Test Malicious Request



403 Forbidden

- WAF inspecting requests
- SQL Injection/XSS protection working

12. Verify in WAF Logs in dashboard

